MANAS [redacted]

[redacted] Science Undergraduate — AI & [redacted] Researcher
Research Focus: Explainable Reinforcement [redacted] for Intelligent Systems
Phone: [redacted]
Email: [redacted]
GitHub: [redacted]
LinkedIn: [redacted]
IELTS: Overall 6.5 (All Bands 6.5)

RESEARCH INTERESTS

Reinforcement [redacted], Explainable Artificial Intelligence (XAI), [redacted] Systems, Intelligent Network Optimization, AI-Native [redacted] Systems

EDUCATION

[redacted] in [redacted] Science
[redacted]
CGPA: 8.33 / 10
Expected Graduation: [redacted]

RESEARCH EXPERIENCE

AI-Native Network Optimization Research

- Designed reinforcement learning-based framework for adaptive resource allocation in intelligent communication networks.
- Investigated optimization strategies for bandwidth prioritization using intent-driven automation.
- Conducted experimental evaluation of intelligent decision-making models for next-generation infrastructure systems.

Explainable [redacted] Systems

- Developed interpretable machine learning models using SHAP-based feature attribution techniques.
- Studied transparency and trust mechanisms in AI-driven decision-making pipelines.

PUBLICATIONS

[redacted], M. "Reevaluating CNN Filter Dimensions: Experimental Findings on CIFAR-10 and Fashion-MNIST"
ICICV 2026 — Accepted for Oral Presentation

[redacted], M. "Intent-Driven AI-Native Network Slicing for Rural Broadcasting over ATSC 3.0/B2X"
WOCC 2026 — Under Review (EDAS)

RESEARCH & INDUSTRY EXPERIENCE

[redacted] Intern 1Stop, [redacted] Jun 2024 – Jul 2024

- Developed supervised learning models (Random Forest, XGBoost) on datasets exceeding 10k records.
- Improved predictive performance through feature engineering and hyperparameter optimization.
- Evaluated models using cross-validation, precision, recall, and statistical performance metrics.

SELECTED RESEARCH PROJECTS

Intent-Driven ATSC Network Slicing (IIT Delhi Winner)

- Proposed reinforcement learning-based adaptive bandwidth allocation framework.
- Designed AI-native telecom slicing architecture for intelligent broadcasting systems.

6G Intelligent Network Management System

- Developed analytics-driven architecture enabling adaptive resource orchestration in next-generation networks.

Explainable AI Loan Prediction System

- Implemented interpretable machine learning framework using SHAP for transparent decision analysis.

AI Traffic Management System

- Developed OpenCV-based real-time traffic congestion detection and density estimation pipeline.

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: Scikit-learn, XGBoost, Ensemble Learning
Deep Learning: [redacted], [redacted], [redacted] Reinforcement Learning
Data Science: Pandas, NumPy, Statistical Analysis, Visualization
Tools: [redacted], [redacted], [redacted], [redacted], Linux
Concepts: Optimization Algorithms, Model Deployment, REST APIs

ACHIEVEMENTS

- 1st Prize — Build-a-thon 3.0, [redacted]
- 1st Prize — Build-a-thon 4.0, [redacted]
- 2nd Prize — AI Agentic Battle

LANGUAGES

Hindi (Native), English (Professional), [redacted] ([redacted])